NutriVision – Literature Review

Project Documentation – Section 2

1. Key Areas Reviewed

- AI-powered nutrition and dietary recommendation systems.
- Personalized recipe generation based on user preferences.
- Prompt Engineering for controlled AI content generation.
- Mobile health and wellness applications.
- Cloud-based storage and multi-device synchronization.
- Responsive mobile user interface design.
- API communication and secure request handling.
- Automated unit and widget testing.

Technology Review

Technology	Role in NutriVision
Flutter	Cross-platform mobile application development from a single codebase.
Artificial Intelligence	Meal analysis, nutritional advice and personalized content generation.
Prompt Engineering	Controls AI output and incorporates dietary restrictions and preferences.
Dio	HTTP communication with external APIs and request/response interception.
Firebase Cloud Firestore	Cloud storage and synchronization of meal records.
Unit Testing	Validation of calculations and application/business logic.
Widget Testing	Validation of Flutter UI components and responsive behavior.

2. Feedback & Evaluation

Evaluation Area	Feedback Focus	Action Taken
Project Idea	Originality, relevance and feasibility.	Refine objectives and scope.
UI/UX	Usability, navigation and visual consistency.	Improve layouts and navigation.
AI Feature	Usefulness and quality of generated recommendations.	Improve prompts and validation.
Technical Implementation	Architecture, code quality and integration.	Refactor and improve modularity.
Testing	Coverage and reliability of test cases.	Add or improve automated tests.
Documentation	Clarity, completeness and academic structure.	Improve explanations and evidence.

3. Suggested Improvements

- Integration with wearable devices.
- Barcode-based food recognition.
- Image-based meal recognition.
- Advanced nutritional analytics.
- Personalized weekly meal plans.
- Integration with professional nutritionists.
- Additional language support.
- Integration with trusted nutritional databases.
- More advanced health and fitness integrations.
- Offline or partially offline intelligent assistance where technically feasible.

4. Final Grading Criteria

Category	Weight
Project Documentation	20%

Requirements and Analysis	10%
UI/UX and Design	10%
Implementation	25%
AI/API/Firebase Integration	10%
Testing	10%
Final Presentation	10%
Innovation and Problem Solving	5%
Total	100%